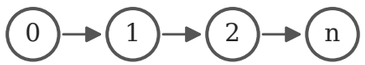
AUGMENTED GAIA: DEPENDENCY-AWARE REFERENCE CONSTRUCTION

Native GAIA task

Text · Document
Vision · Audio



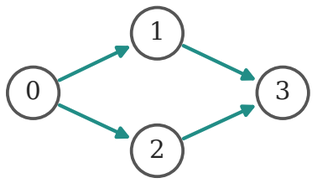
165 tasks · native chains

AD/SD annotation

LLM witnesses +
deterministic compiler

1,103 AD + 231 SD
1,334 semantic records

Dependency DAG



1,284 nodes
1,207 reduced edges

Recorded orderings

[0, 1, 2, 3] Native
[0, 2, 1, 3] candidate
…

1,786 non-native candidates
72 tasks with alternatives

Main reference set

165 Native + 1,786 candidates
1,951 references in total

Full unfiltered pool
63 multi-order-rich tasks

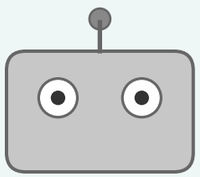
Tool layer: 379 gold slots / 144 tasks

14 actions (open) · 16 (closed)

Replay + anonymous human AD/SD review provide separate quality evidence; they do not filter the main reference set.

Max over Native and every candidate; Native wins ties

STAGED INFERENCE · THREE-FACET EVALUATION



Agent under test

Task + shared
action catalog

(A) Plan structural fidelity

S3 pre-execution graph vs. best reference

NodeF1 · EdgeF1

PlanningScore = (NodeF1 + EdgeF1) / 2

(B) Tool invocation accuracy

Realized calls vs. gold tool slots

ToolF1 · ParamF1 · ParamValueF1

ToolUsageScore = mean of the three

(C) Final-answer correctness

Returned response vs. reference answer

Norm. EM · TokenF1 · Accepted

Headline: human Accepted Answer rate

Planning · S1–S3

Abstract plan → tool assessment
→ concrete pre-execution plan

Execution · S4

Tool calls ↔ observations
Actual arguments and outputs retained

Returned answer

Open: synthesized S5 response
Closed: submission / final fallback

S3 is primary; S1 is auxiliary. Predictions and returned answers stay fixed when planning references change.